Worksheet 2.5 — The Quadratic Formula and Discriminant

Mathematics: Analysis & Approaches SL (IB) · Unit 2: Functions

The Big Ideas

The discriminant predicts how many real solutions a quadratic has before you finish solving it.

- $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- $\Delta = b^2 - 4ac$: $\Delta > 0$ two real roots, $\Delta = 0$ one repeated root, $\Delta < 0$ no real roots.

Worked Examples

Example 1 — Solve with the formula

Solve $3x^2 - x - 4 = 0$.

Solution. $a = 3, b = -1, c = -4$: $\Delta = 1 + 48 = 49$. $x = \dfrac{1 \pm 7}{6}$.

Conclusion: $x = \dfrac{4}{3}$ or $x = -1$.

Example 2 — Nature of the roots

Determine the nature of the roots of $2x^2 + 3x + 5 = 0$.

Solution. $\Delta = 9 - 40 = -31 < 0$.

Conclusion: no real roots.

Example 3 — Repeated root condition

Find k so that $x^2 - 8x + k = 0$ has a repeated root.

Solution. $\Delta = 0$: $64 - 4k = 0 \Rightarrow k = 16$.

Conclusion: $k = 16$.

Example 4 — Application

A ball's height is $h(t) = -5t^2 + 18t + 1$. Determine whether it ever reaches 20 m.

Solution. Set $-5t^2 + 18t + 1 = 20 \Rightarrow -5t^2 + 18t - 19 = 0$. $\Delta = 324 - 380 = -56 < 0$.

Conclusion: no real solution, so it never reaches 20 m.

Example 5 — Solving using the discriminant to predict, then confirm

Show that $x^2 - 6x + 9 = 0$ has a repeated root, and find it.

Solution. $\Delta = 36 - 36 = 0$, confirming a repeated root. $x = \dfrac{6 \pm 0}{2} = 3$.

Conclusion: repeated root at $x = 3$.

Example 6 — Two-parameter repeated-root condition

Find both values of m so that $x^2 + mx + 25 = 0$ has a repeated root.

Solution. $\Delta = 0$: $m^2 - 100 = 0 \Rightarrow m = \pm 10$.

Conclusion: $m = 10$ or $m = -10$.

Example 7 — Discriminant with a parameter inside a real scenario

A rectangle's area is $x^2 + kx + 9 = 25$ for side length x. Find the value(s) of k for which exactly one value of x solves this.

Solution. Rearranged: $x^2 + kx - 16 = 0$. Exactly one solution requires $\Delta = 0$: $k^2 + 64 = 0$ — impossible for real k (since $k^2 \geq 0$, $k^2 + 64 > 0$ always).

Conclusion: no real value of k gives exactly one solution — this equation always has two distinct real roots (one positive, one negative, since the constant term is negative).

Example 8 — Comparing two quadratics via discriminants

Determine which of $x^2 - 4x + 5 = 0$ and $x^2 - 4x + 3 = 0$ has real roots.

Solution. First: $\Delta = 16 - 20 = -4 < 0$ (no real roots). Second: $\Delta = 16 - 12 = 4 > 0$ (two real roots).

Conclusion: only the second equation has real roots.

Example 9 — Application — break-even analysis

A company's profit is $P(x) = -2x^2 + 40x - 150$ (in \$1000s), for x hundred units sold. Determine, using the discriminant, whether the company can ever break even ($P = 0$).

Solution. $\Delta = 1600 - 1200 = 400 > 0$.

Conclusion: yes — two real break-even points exist (solve $-2x^2 + 40x - 150 = 0$ to find $x = 5$ or $x = 15$).

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve $x^2 - 3x - 10 = 0$ using the quadratic formula.

Question 2

Find the discriminant of $4x^2 + 4x + 1 = 0$ and describe the roots.

Question 3

Find the discriminant of $x^2 + 2x + 6 = 0$ and describe the roots.

Question 4

Find $k > 0$ so that $2x^2 + kx + 8 = 0$ has a repeated root.

Question 5

For what values of k does $x^2 - 4x + k = 0$ have two distinct real roots?

Question 6

Solve $5x^2 + 2x - 3 = 0$ using the quadratic formula.

Question 7

A rectangle has perimeter 40 m and area 91 m^2; set up and solve a quadratic to find its dimensions.

Question 8

Find both values of m so that $x^2 + mx + 36 = 0$ has a repeated root.

Question 9

Determine which of $2x^2 + 3x + 7 = 0$ and $2x^2 + 9x + 7 = 0$ has real roots.

Question 10

A company's profit is $P(x) = -x^2 + 18x - 32$ (\$1000s); determine using the discriminant whether it can break even, then find the break-even points.

Question 11 — Challenge

Find the value(s) of m so that the line $y = mx$ is tangent to the parabola $y = x^2 + 4$ (hint: set them equal and require $\Delta = 0$).

Question 12 — Challenge

Find the value(s) of k so that the line $y = 2x + k$ is tangent to $y = x^2$.

Question 13 — Challenge

For what values of k does $kx^2 - 4x + 1 = 0$ (with $k \neq 0$) have no real roots?

Question Recap & Answer Key

Question Recap

1. Solve $x^2 - 3x - 10 = 0$ using the quadratic formula.
2. Find the discriminant of $4x^2+4x+1 = 0$ and describe the roots.
3. Find the discriminant of $x^2+2x+6 = 0$ and describe the roots.
4. Find $k > 0$ so that $2x^2 + kx + 8 = 0$ has a repeated root.
5. For what values of k does $x^2 - 4x + k = 0$ have two distinct real roots?
6. Solve $5x^2 + 2x - 3 = 0$ using the quadratic formula.
7. A rectangle has perimeter 40 m and area 91 m^2; set up and solve a quadratic to find its dimensions.
8. Find both values of m so that $x^2 + mx + 36 = 0$ has a repeated root.
9. Determine which of $2x^2+3x+7 = 0$ and $2x^2+9x+7 = 0$ has real roots.
10. A company's profit is $P(x) = -x^2 + 18x - 32$ (\$1000s); determine using the discriminant whether it can break even, then find the break-even points.
11. Find the value(s) of m so that the line $y = mx$ is tangent to the parabola $y = x^2 + 4$ (hint: set them equal and require $\Delta = 0$).
12. Find the value(s) of k so that the line $y = 2x + k$ is tangent to $y = x^2$.
13. For what values of k does $kx^2 - 4x + 1 = 0$ (with $k \neq 0$) have no real roots?

✓ Answer Key

1. $x = \dfrac{3 \pm 7}{2}$: $x = 5$ or $x = -2$
2. $\Delta = 0$: one repeated root ($x = -\frac{1}{2}$)
3. $\Delta = 4 - 24 = -20 < 0$: no real roots
4. $k^2 = 64 \Rightarrow k = 8$
5. $16 - 4k > 0 \Rightarrow k < 4$
6. $\Delta = 4 + 60 = 64$; $x = \dfrac{-2 \pm 8}{10}$: $x = \frac{3}{5}$ or $x = -1$
7. $x + y = 20$, $xy = 91$; $x^2 - 20x + 91 = 0 \Rightarrow (x - 7)(x - 13) = 0$: dimensions 7 m by 13 m
8. $m^2 = 144 \Rightarrow m = \pm 12$
9. first: $\Delta = 9 - 56 = -47 < 0$ (no real roots); second: $\Delta = 81 - 56 = 25 > 0$ (real roots) — only the second has real roots
10. $\Delta = 324 - 128 = 196 > 0$: yes; $x = \dfrac{18 \pm 14}{2}$: $x = 16$ or $x = 2$
11. $x^2 + 4 = mx \Rightarrow x^2 - mx + 4 = 0$; $\Delta = m^2 - 16 = 0 \Rightarrow m = \pm 4$
12. $x^2 = 2x + k \Rightarrow x^2 - 2x - k = 0$; $\Delta = 4 + 4k = 0 \Rightarrow k = -1$
13. $\Delta = 16 - 4k < 0 \Rightarrow 16 < 4k \Rightarrow k > 4$